

Relationship Between Education and Earned Income

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Introduction:

According to educationdata.org over 18.6 million individuals in the United States enrolled in college in 2022, of which 3.2 million are pursuing graduate degrees. While at the same time, the average cost of college in the United States has risen to \$38,270 per student per year. Given the rising costs and implications for individual decision making and public policy, it is more important than ever to accurately evaluate the long-term effects education has on projected annual income.

By utilizing Current Population Survey (CPS) data, this report aims to provide a detailed analysis of the factors which contribute to earned income with a specific focus on the effects of higher education. Additionally, statistical tests are performed to determine the significance of these factors in predicting annual income.

Data Used

The CPS is administered monthly by the Census Bureau to around 60,000 households. Demographic, education, and labor, household, and regional data are collected and compiled into individual and household level datasets. To account for demographic and regional effects, both datasets are merged for comprehensive analysis. Variables indicative of age, gender, citizenship status, health, ethnicity, marital status, region, and level of education are evaluated to predict their impact on earned income. CPS indicator variables for missing or imputed data are also utilized to assess the impact of missing data on the model. The reliability, size, and comprehensiveness of the CPS makes it an ideal data source for this analysis.

Analysis Sample

Table 1: Data Summary

<i>Variable</i>	<i>Mean</i>	<i>Std Dev</i>	<i>Minimum</i>	<i>Maximum</i>
<i>Citizenship Status (1 for Citizen)</i>	0.902	0.297	0	1
<i>Binary Sex Indicator (1 for Female)</i>	0.465	0.499	0	1
<i>Poor Health Indicator (1 for Poor Health)</i>	0.066	0.249	0	1
<i>Married</i>	0.718	0.45	0	1
<i>Widowed</i>	0.01	0.1	0	1
<i>Divorced</i>	0.099	0.298	0	1
<i>Separated</i>	0.026	0.158	0	1
<i>Annual Earned Income</i>	57706.74	44621.48	1208	296000
<i>Annual Hours Worked</i>	2021.037	591.426	10	4160
<i>Hourly Wage</i>	28.241	20.313	5.001	192.308
<i>Hispanic</i>	0.198	0.399	0	1
<i>White</i>	0.609	0.488	0	1
<i>Black</i>	0.105	0.306	0	1
<i>Asian</i>	0.067	0.251	0	1
<i>Other</i>	0.021	0.143	0	1
<i>High School Diploma</i>	0.245	0.43	0	1
<i>Some College</i>	0.289	0.453	0	1
<i>Bachelor's Degree</i>	0.246	0.431	0	1
<i>Master's Degree</i>	0.112	0.316	0	1
<i>Professional or PhD</i>	0.032	0.177	0	1
<i>Education Categorical Indicator</i>	3.173	1.225	1	6
<i>Missing Earnings Data Indicator</i>	0.242	0.428	0	1
<i>Missing Demographics Data Indicator</i>	0.338	0.473	0	1
<i>South</i>	0.36	0.48	0	1
<i>Northeast</i>	0.157	0.364	0	1
<i>Midwest</i>	0.198	0.399	0	1
<i>West</i>	0.285	0.452	0	1
<i>Median State Income</i>	46800.58	6351.908	35501	75000
<i>Age in Years</i>	41.841	10.037	19	60
<i>Age Squared / 100</i>	18.514	8.421	3.61	36
<i>At Least High School Diploma Indicator</i>	0.925	0.263	0	1
<i>At Least Bachelor's Degree Indicator</i>	0.391	0.488	0	1
<i>Categorical Marital Status Variable</i>	1.877	1.496	1	5
<i>Years of Education - 12</i>	2.076	2.733	-11	8
<i>Age - 18</i>	23.841	10.037	1	42

Analysis Sample Continued

Participant education levels range from less than a high school diploma to advanced degrees. Roughly 24.5% of individuals reported having only a high school diploma, 28.9% had some college experience, and 24.6% held a bachelor's degree. Graduate degrees were less common, with 11.2% earning a master's and 3.2% achieving a professional or PhD degree. 92.5% of the sampled individuals have at least a high school diploma and 39.1% have at least a bachelor's degree. On average, individuals attain roughly 2 additional years of education beyond the 12 that traditionally comprise a high school diploma.

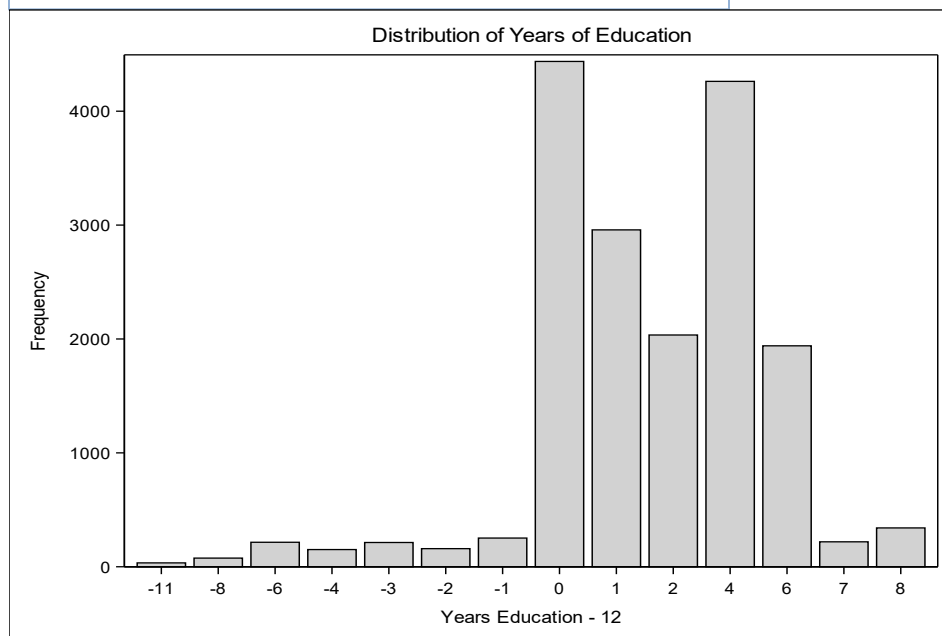
Annual earned income averaged \$57,706.74, with a wide standard deviation of \$44,621.48. The lowest reported income was \$1,208, while the highest was \$296,000. Sampled individuals worked an average of 2,021 hours per year with an average hourly wage of \$28.24. Hourly wages ranged from \$5.00 to \$192.31.

The sample's average age was 41.84 years, with a range from 19 to 60. Most participants identify as white (60.9%), followed by Hispanic (19.8%), Black (10.5%), Asian (6.7%), and Other (2.1%). About 90.2% of the sample are citizens. Only 6.6% of individuals report poor health. In terms of marital status, 71.8% are married, 9.9% are divorced, 2.6% are separated, and 1.0% are widowed while 14.7% have never married.

Sampled individuals prominently reside in the South (36%), followed by the West (28.5%), Midwest (19.8%), and Northeast (15.7%).

Distribution of Key Variables

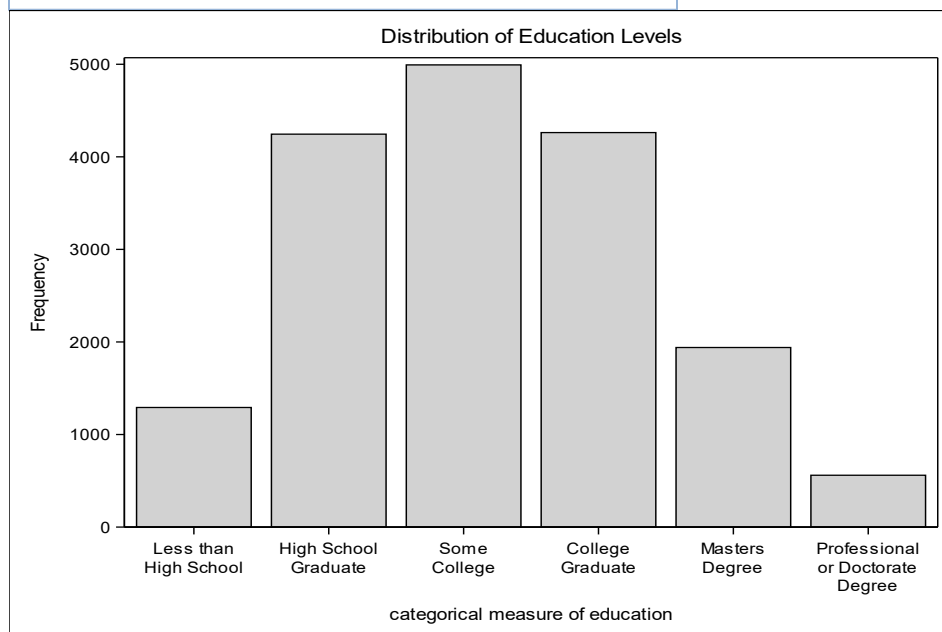
Figure 1: Years of Education Frequency Distribution



The 'Years of Education' variable has been adjusted to represent the total years of education beyond high school, with a value of 0 corresponding to 12 years of education (a high school diploma). The distribution reveals that the largest group consists of individuals

with 12 years of education, followed closely by those with 16 years (4 years beyond high school), equivalent to a bachelor's degree. The majority of sampled individuals fall within the range of 12 to 18 years of education.

Figure 2: Education Level Frequency Distribution

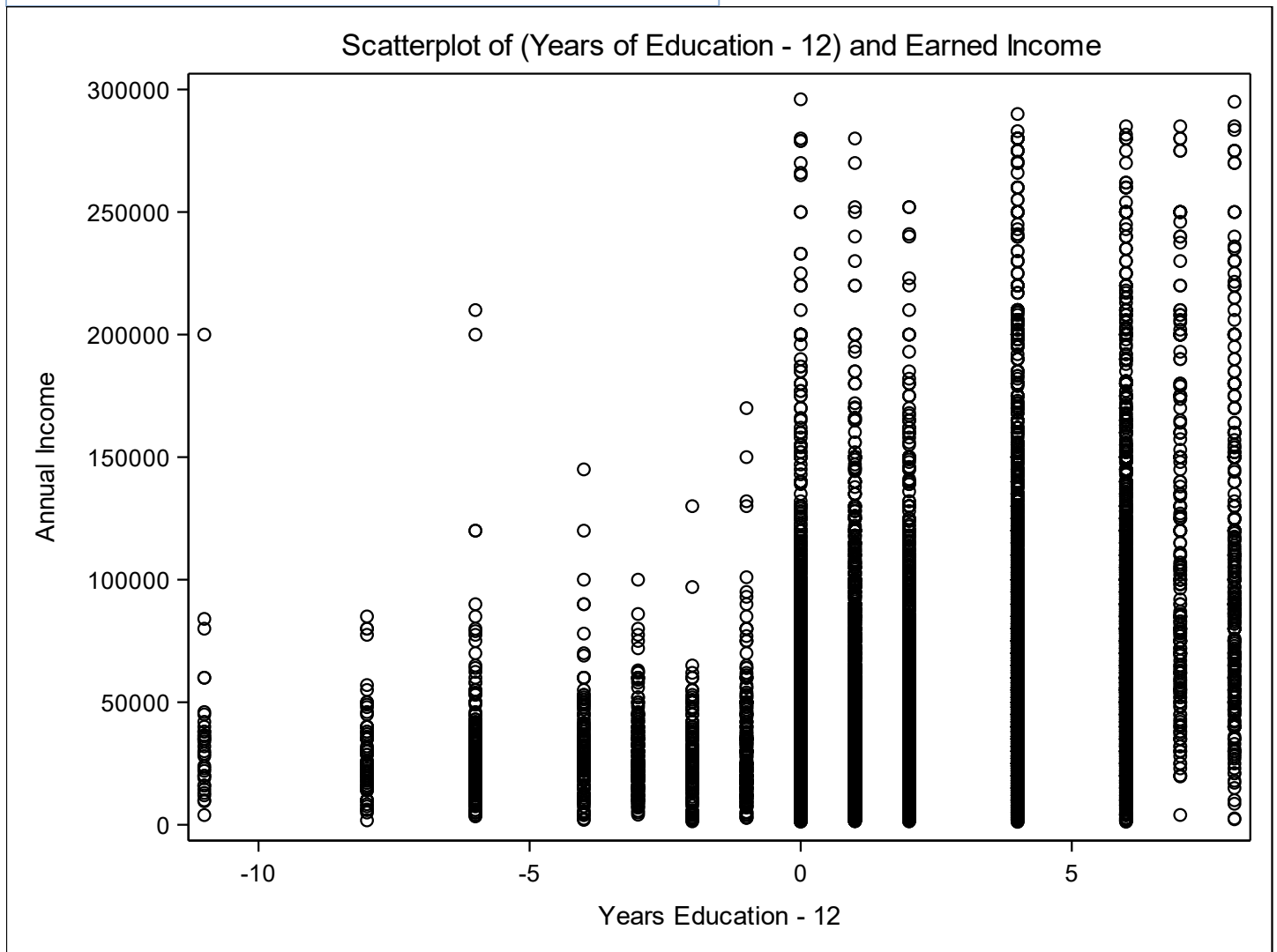


The distribution of education levels echoes the concentrations exhibited above in years of education. While some individuals sampled have less than a high school education, the majority either have a high school diploma, some college, or a bachelor's degree. There is a sharp

drop observed from bachelor's to master's degrees with the number of individuals represented more than halved. This trend continues for master's to PhD/Professional – the level with the lowest frequency.

Distribution of Key Variables Continued

Figure 3: Observed Incomes at Various Years of Education



An upward trend in the scatterplot shows that higher levels of education are associated with observed increases in earned income. Zero on the x-axis refers to individuals with 12 years of education, generally, a high school diploma. As evidenced by the scatterplot, as years of education increase to 4+ years beyond high school, the concentration of individuals nearest the bottom of annual income decreases. However, there remains a high degree of variability across all education levels. Some individuals with less than a high school diploma still earn relatively high incomes, some with advanced degrees, relatively low. Though, the proportion of individuals earning more than \$100k annually becomes much greater as years of education increase.

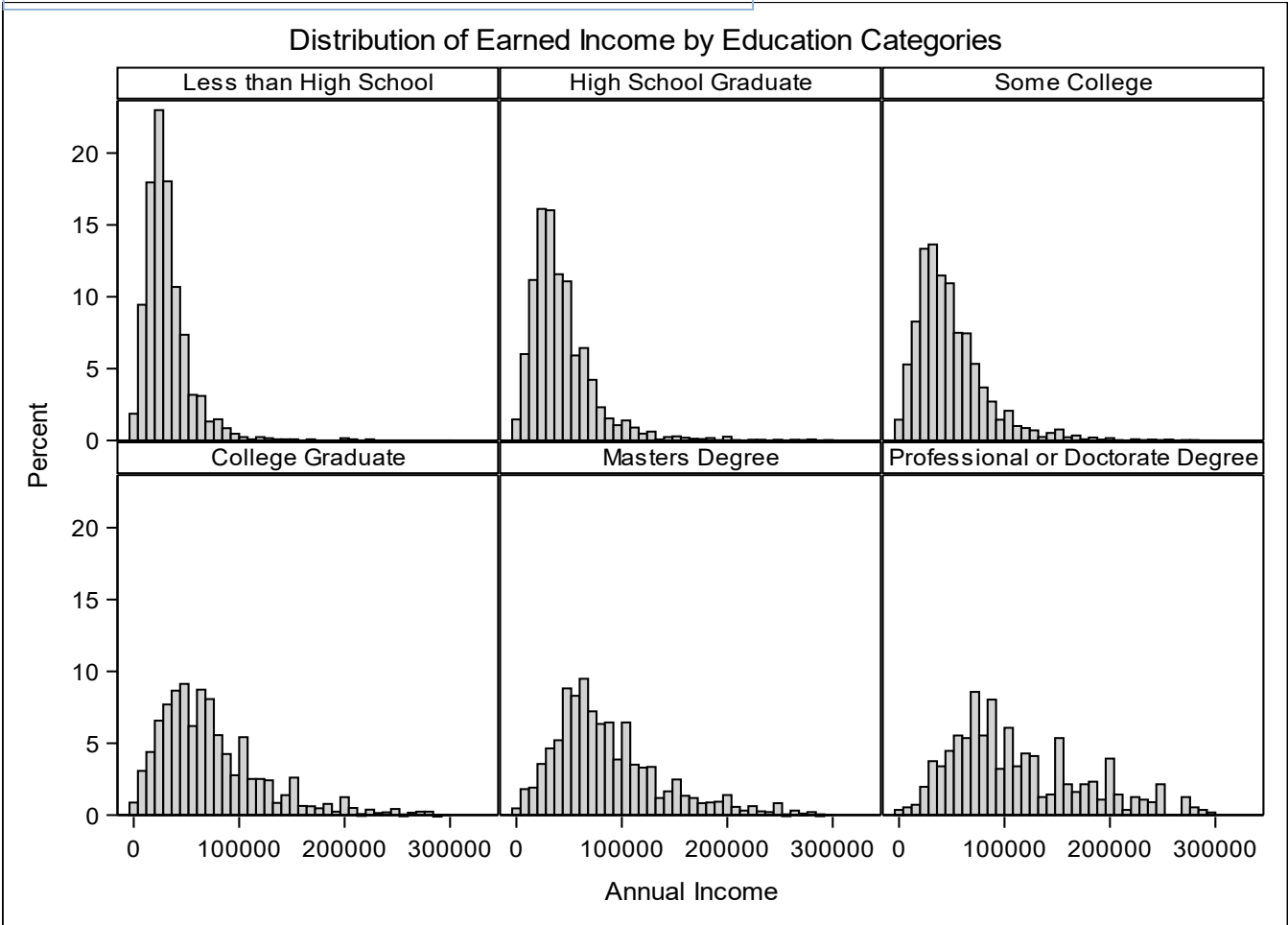
Bivariate Relationships Between Education and Earnings

Examining the means and distributions of earned income by education category provides a clearer picture of the relationship between these variables. The rising mean incomes represented in Table 2 can be seen visualized in the Figure 4 distributions below. As educational attainment increases, not only do the means for earned income rise, but also large increases in variability can be seen where concentrations of earnings under \$50,000 are dispersed more evenly among a wider range of income levels. This is evidenced by the evolution of the widening distribution curves as for higher levels of education.

Table 2: Education Level Mean Incomes

Education Level	Mean
<i>Less than High School</i>	\$31,126.20
<i>High School Diploma</i>	\$42,353.25
<i>Some College</i>	\$48,106.43
<i>Bachelor's Degree</i>	\$72,568.86
<i>Master's Degree</i>	\$85,546.69
<i>Professional or PhD</i>	\$111,456.99

Figure 4: Earned Income Frequencies by Education Levels



Regression Analysis

Regression Model 1: Single Variable Analysis

Table 3: Regression on Years of Educations	Estimated Impact (in \$)	Margin of Error (SE)	Test Statistic (t)	Significance (p-value)
<i>Baseline Income (12 Years of Education)</i>	43,993.00	389.69	112.89	< 0.0001
<i>Additional Year(s) of Education Beyond High School</i>	6,606.80	113.56	58.18	< 0.0001

Using a simple linear regression model, this analysis examines the relationship between years of education and income earned annually. The model explains 16.37% of the variation in income, implying that years of education is a meaningful predictor, yet outside factors also contribute significantly to income variation. The model predicts a baseline income for individuals with 12 years of education of \$43,993. For each additional year of education, an individual is predicted to earn, on average, \$6,606.80 more annually. Both the model and its predictors are statistically significant ($p < 0.0001$). These findings highlight education's role in income levels while suggesting other factors should also be explored.

Regression Model 2: Multiple Variable Analysis

The following model incorporates age, ethnicity, health, marital, and citizenship status, as well as regional data to examine other factors influencing income. Median income by state is also included to account for regional cost-of-living variations. Observations with imputed data are omitted. This approach allows for a more precise evaluation of education's effect on income. The intercept represents the baseline for all excluded or zeroed variables, corresponding to an 18-year-old white male, who has never been married, did not complete high school, and resides in the South. The column 'Estimated Impact (in \$)' is the predicted change in annual income given a different value than the intercept in the associated variable, holding all else equal.

Regression Model 2: Multiple Variable Analysis Continued

Table 4: Multi-Variable Regression

	Estimated Impact (in \$)	Margin of Error (SE)	Test Statistic (t)	Significance (p-value)
<i>Intercept</i>	-8154.91	3680.78	-2.22	0.0267
<i>Age - 18 (additional year(s) over 18)</i>	520.11	37.94	13.71	<.0001
<i>Binary Sex Indicator (1 for Female)</i>	-25053.00	740.40	-33.84	<.0001
<i>Citizenship Status (1 for Citizen)</i>	5147.94	1521.88	3.38	0.0007
<i>Poor Health Indicator (1 for Poor Health)</i>	-10233.00	1493.47	-6.85	<.0001
<i>Hispanic</i>	-6623.16	1146.73	-5.78	<.0001
<i>Black</i>	-4369.46	1261.73	-3.46	0.0005
<i>Asian</i>	-112.60	1498.92	-0.08	0.9401
<i>Other</i>	-8005.87	2415.78	-3.31	0.0009
<i>Married</i>	6724.04	1170.64	5.74	<.0001
<i>Widowed</i>	-58.78	3633.02	-0.02	0.9871
<i>Divorced</i>	5522.16	1530.46	3.61	0.0003
<i>Separated</i>	4482.64	2466.25	1.82	0.0692
<i>High School Diploma</i>	8412.62	1704.47	4.94	<.0001
<i>Some College</i>	15214.00	1691.66	8.99	<.0001
<i>Bachelor's Degree</i>	35900.00	1732.88	20.72	<.0001
<i>Master's Degree</i>	48148.00	1885.06	25.54	<.0001
<i>Professional or PhD</i>	70360.00	2429.30	28.96	<.0001
<i>Northeast</i>	-485.98	1196.40	-0.41	0.6846
<i>Midwest</i>	-1095.53	989.21	-1.11	0.2681
<i>West</i>	4166.03	917.54	4.54	<.0001
<i>Median State Income</i>	0.72	0.06	11.57	<.0001

Looking specifically at education in the above table, the model predicts that, holding all else equal, an individual who earns a high school diploma adds \$8,412.62 to their annual income. As education levels increase, so do these predicted values, with Some College adding \$15,214.00, a Bachelor's \$35,900.00, Master's \$48,148.00, and a PhD or Professional degree adding \$70,360.00. These education level indicators are statistically significant (all p-values <.0001) and indicate that, even when other factors are accounted for, education remains a significant factor in predicting earned income. The model explains 32.15% of the variation in income – an improvement over the simple regression. Still, this implies additional outside factors also contribute significantly to income variation.

Regression Model 2: Multiple Variable Analysis Continued

Gender disparities are also evident, with the model showing a significant negative coefficient for females, pointing toward a wage gap unexplained by other variables in the model. Poor health, ethnicity, and marital status further illustrate structural factors influencing income, with married individuals earning significantly more on average. Finally, regional differences and the positive association with median state income suggest that factors, such as the nature of local economies and cost of living in a given area, play a crucial role in shaping individual income.

Missing Data and Imputation Sensitivity Analysis

Table 5: Cross Tabulation of Imputed Data

In this section, we analyze the degree and effect of missing data in the sample. The table looks at the relationship between missing earnings and missing demographic

	Complete earnings data	Incomplete earnings data	Total
Complete demographic data	11228 <i>64.93%</i>	213 <i>1.23%</i>	11441 <i>66.16%</i>
Incomplete demographic data	1876 <i>10.85%</i>	3975 <i>22.99%</i>	5851 <i>33.84%</i>
<i>Total</i>	13104 <i>75.78%</i>	4188 <i>24.22%</i>	17292 <i>100%</i>

data. The first row is for observations with complete demographic information. In this case, only 1.23 percent of observations with complete demographic data have incomplete earnings data. The second row shows that 22.99% of the observations with incomplete demographic data also have incomplete earnings data. To evaluate whether data imputation has significant effects on the estimates of the model, we filter for only those observations with complete data and contrast against observations with incomplete data. The following table calculates the percentage change from modeling complete to incomplete data, indicates whether the sign of the parameter has changed, and indicates changes in statistical significance at a 95% confidence level.

Missing Data and Imputation Sensitivity Analysis Continued

Table 6: Multi-Variable Regression (imputed data)	Parameter Estimates (Complete Data)	Parameter Estimates (Incomplete Data)	% Change from Complete to Incomplete Data	Sign Change?	Change in Statistical Significance?
<i>Intercept</i>	-8154.91	9981.07	222%	Yes	Yes
<i>Age – 18 (additional year(s) over 18)</i>	520.11	530.64	-2%	No	No
<i>Binary Sex Indicator (1 for female)</i>	-25053.00	-21735.00	13%	No	No
<i>Citizenship Status(1 for citizen)</i>	5147.94	2711.32	-47%	No	Yes
<i>Poor Health Indicator (1 for poor health)</i>	-10233.00	-7418.12	28%	No	No
<i>Hispanic</i>	-6623.16	-4550.30	31%	No	No
<i>Black</i>	-4369.46	-6443.16	-47%	No	No
<i>Asian</i>	-112.60	-455.28	-304%	No	No
<i>Other</i>	-8005.87	-7914.21	1%	No	Yes
<i>Married</i>	6724.04	5712.89	-15%	No	No
<i>Widowed</i>	-58.78	-923.32	-1471%	No	No
<i>Divorced</i>	5522.16	1568.12	-72%	No	Yes
<i>Separated</i>	4482.64	-256.86	-106%	Yes	No
<i>High School Diploma</i>	8412.62	9845.49	17%	No	No
<i>Some College</i>	15214.00	17181.00	13%	No	No
<i>Bachelor's Degree</i>	35900.00	38804.00	8%	No	No
<i>Master's Degree</i>	48148.00	49104.00	2%	No	No
<i>Professional or PhD</i>	70360.00	79104.00	12%	No	No
<i>Northeast</i>	-485.98	-5594.37	-1051%	No	Yes
<i>Midwest</i>	-1095.53	205.21	119%	Yes	No
<i>West</i>	4166.03	-1850.10	-144%	Yes	Yes
<i>Median State Income</i>	0.72	0.38	47%	No	No

The model comparison (Table 6) above shows multiple differences and consistencies in parameter estimates between complete and imputed data. Education variables maintain consistent relationships with earned income across both models. Education level indicators demonstrate relatively small percentage changes – ranging from 2% to 17%, with no changes in sign or statistical significance. Demographic variables are much more affected by the imputation of data. For example, "Widowed" shows -1471% change without a sign reversal with regional variables like "Northeast" and "West" also showing substantial differences, with the

"Northeast" parameter changing by -1051% and "West" by -144%, both accompanied by shifts in statistical significance. Additionally, variables such as "Citizenship Status" and "Divorced" display notable percentage changes of -47% and -72%, respectively, also changing in their statistical significance. This comparison highlights education level as a significant predictor of earned income, in both the complete and imputed data.

Key Takeaways

The analysis indicates a clear and statistically significant relationship between education level and annual earnings. At all levels, the model predicts that, holding all else equal, higher educational attainment leads to higher income. For example, individuals with less than a high school education report an average income of \$31,126, while those with advanced degrees, such as a professional or doctoral degree, earn around \$111,457 on average – a difference of over \$80,000. The single variable regression model also predicts that for each additional year of education beyond high school the expected annual income increases by \$6,606.80.

However, large standard errors and wide income ranges within the education categories highlight that other unaccounted-for factors (such as personal motivation or social connections) may also play a role.

The inclusion of demographic variables in the model allows for deeper look into income disparities. Age, sex, citizenship status, ethnicity, relationship status, and location all play a significant role in predicting earned income. Nevertheless, educational attainment is second only to sex in its significance as a factor affecting the predicted value of earned income. The analysis confirms a well-known gender pay gap, but also emphasizes the importance of education in increasing economic mobility. This insight can not only guide individual academic goals but also policy efforts aimed at improving social equity through access to higher education.